

Academic performance and dental student satisfaction with emergency remote teaching of endodontics during COVID-19 pandemic: A retrospective cohort study

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Submission: 09-02-21

Revision: 22-03-21

Acceptance: 25-03-21

Web publication: 03-09-21

INTRODUCTION

In March 2020, the World Health Organization declared the coronavirus disease 2019 (COVID-19) outbreak as a

pandemic and released a list of recommendations which included social distancing.^[1] The mode of transmission

Access this article online	
Quick Response Code:	Website: www.saudiendodj.com
	DOI: 10.4103/sej.sej_30_21

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How to cite this article: Alkahtani FN, Almohareb RA, Barakat RM. Academic performance and dental student satisfaction with emergency remote teaching of endodontics during COVID-19 pandemic: A retrospective cohort study. Saudi Endod J 2021;11:321-6.

was not determined and asymptomatic patients were considered as a possible source of carrying and transmitting the infection.^[2-4] COVID-19 was considered an international crisis that caused an interruption of typical activities such as on-campus education.^[5-8] Therefore, all academic plans were required to be provided online.^[9]

The entire shift to remote teaching was not possible in clinical and preclinical training in dentistry.^[10-13] The undergraduate endodontic didactic component was provided online for the first time in the dental college of Princess Nourah Bint Abdulrahman University (PNU), Riyadh, Saudi Arabia. The dental students and full-time faculty adopted the online approach in education within 10 days of campus lockdown. This type of urgent and unplanned transferable in the mode of course delivery was regarded as an emergency remote teaching rather than a typical online education.^[14] The online education composure requires comprehensive planning of the online course concerning modality, pacing, and form of assessment.^[14] The role of the instructor and students is carefully studied and their type of interaction and collective feedback will be determined before the beginning of the online course.^[14] However, in emergency remote teaching, the faculty and students were not prepared for remote teaching and were required to adapt to the current circumstances urgently.^[14] The preclinical endodontic course is a critical component of undergraduate dental education, as it provides students with fundamental lifelong skills in clinical diagnosis and management. The impact of online modalities used in emergency remote teaching on dental education needs further support in the literature. A recent cross-sectional study suggested that students and lecturers had a positive perspective of online learning used during COVID-19.^[15] Moreover, lecture attendance was not correlated with academic performance in a clinical endodontic course taken in Saudi dental school.^[16] Another study found a weak association between the didactic academic performance and preclinical skills which may advocate the use of online modalities for teaching the didactic component of endodontic courses in emergency remote teaching.^[17]

Therefore, this study aimed to evaluate the consequences of introducing online modalities used in emergency remote teaching on students learning satisfaction. Moreover, it sought to compare these students' academic performance in the summative examination with that of their colleagues who were taught face to face in the prepandemic year.

MATERIALS AND METHODS

Study design and subjects

This retrospective cohort study was conducted at Princess Nourah Bint Abdulrahman University; College of Dentistry (PNU-CD), after ethical approval was obtained from PNU Institutional Review Board (IRB # 20-0293).

The cohort was 39 undergraduate students at PNU-CD, who attended the preclinical endodontic course during the 2019–2020 academic years. The module (block) course given to 3rd-year dental students consists of a didactic component designed to provide students with a thorough knowledge of the fundamental principles of successful endodontic treatment, in addition to hands-on training in the simulation laboratory. The main teaching method of the didactic component is direct instruction through conventional face-to-face lectures. During the 2019–2020 academic years, due to the COVID-19 pandemic, this was no longer possible, and emergency remote teaching was adopted. Lectures had to be given online either in the form of online-synchronized lectures (virtual lectures) using Microsoft Teams platform (Microsoft Corporation, Redmond, Washington, United States.) or asynchronized (recorded) lectures posted on Blackboard LMS (Blackboard Inc., Washington, District of Columbia, United States.). The preclinical endodontic course was composed of nineteen lectures of which four were face-to-face lectures, eleven were online-synchronized, and four were online-asynchronized lectures.

Survey of student satisfaction with methods of lecture delivery

After all the lectures were delivered, and before taking their summative assessments, students were invited via E-mail to fill an online questionnaire regarding their satisfaction with the three methods of lecture delivery. Students were asked to rate their satisfaction according to three categories: satisfied, neutral, or unsatisfied. The first question inquired about students' overall satisfaction with online teaching during the course. The second question was about their preferences regarding the best method of lecture delivery used during the course. This was followed by a set of 18 questions asking the students to rate their satisfaction with the three methods used for lecture delivery in terms of comprehension and retention of knowledge, communication with the lecturer, stress-free learning, time convenience, Internet, and technical support in online modalities, and transportation

for face-to-face teaching. The purpose of the survey was explained beforehand, and the voluntary and anonymity of participation were emphasized.

Student performance

Assessment of knowledge in the preclinical endodontic course is mainly conducted through two summative examinations, which consist of multiple-choice questions placed according to a fixed blueprint defining the number of questions per lecture. The questions must conform to specific guidelines monitored by the PNU-CD Assessment Unit. The ExamSoft platform (ExamSoft Worldwide Inc., Boca Raton, Florida, United States.) through which summative examinations were carried out also allows item analysis of student performance. On completion of the exams, students' overall scores were calculated, then compared to scores of dental students in the prepandemic preclinical endodontic course. Lectures and lecturers during the two consecutive academic years were identical and all 37 students enrolled in the course of the previous year had successfully passed.

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CONCLUSION

Within the limitation of this study, online modalities used in emergency remote teaching were as satisfactory as conventional face-to-face teaching in the preclinical endodontic course. The shift from conventional face-to-face to emergency remote teaching did not affect students' didactic knowledge as measured by their summative examination performance. Furthermore, students significantly favored online-synchronized lectures because this modality allowed for better communication and interaction with the lecturer. However, students were unsatisfied with the time allocated for online-synchronized lectures during emergency remote teaching. Further research is recommended to explore and expand the role of online teaching in dental education.

Financial support and sponsorship

This research was funded by the Deanship of Scientific Research at Princess Nourah bint Abdulrahman University through the Fast-track Research Funding Program.

Conflicts of interest

There are no conflicts of interest.

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